HW #5

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March 5, 2026

1.

(a) The "primitive soup" hypothesis says life began in a the ocean where UV light and lightning produced organic compounds like amino acids and nucleotides. The "soup" has no way to ensure that polymers form with the correct chemical bonds. Proteins require peptide bonds and RNA requires specific nucleotide linkages, but Miller's experiments produce a mix of bonds, which prevent replication entirely.

The "primitive pizza" hypothesis says life began on positively charged iron pyrite surfaces. Organic ions bond to the surface and are held in a fixed orientation, able to move only in two dimensions instead of freely in solution, increasing the reaction speed. This directly solves the soup's problem because the surface enforces correct orientations for polymerization.

(b) Without replicase enzymes the error rate is very high, too high to copy any meaningful genetic sequence reliably. With replicase but without proofreading, the error rate drops to 1/1000 to 1/10,000 per nucleotide. RNA strands separate immediately after copying so there is no way to check for mistakes, unlike DNA which uses the second strand to catch errors. Replicase helps but proofreading is what really makes large genomes viable.

(c) RNA is single-stranded so errors cannot be proofread. With a human genome of 3 billion bases and an error rate of 1/1000 to 1/10,000, every replication cycle would introduce thousands to millions of mutations, so no useful genetic information would survive.

2.

(a)

$$m_{\text{proton}} = m_{\text{neutron}} = 1.67 \times 10^{-27} kg$$
$$m_{^{12}\text{C}} = 12 \times 1.67 \times 10^{-27} \text{ kg} = 2.004 \times 10^{-26} \text{ kg}$$
$$m_{^{13}\text{C}} = 13 \times 1.67 \times 10^{-27} \text{ kg} = 2.171 \times 10^{-26} \text{ kg}$$

(b)

$$E_1 = \frac{h^2}{8mL^2}$$

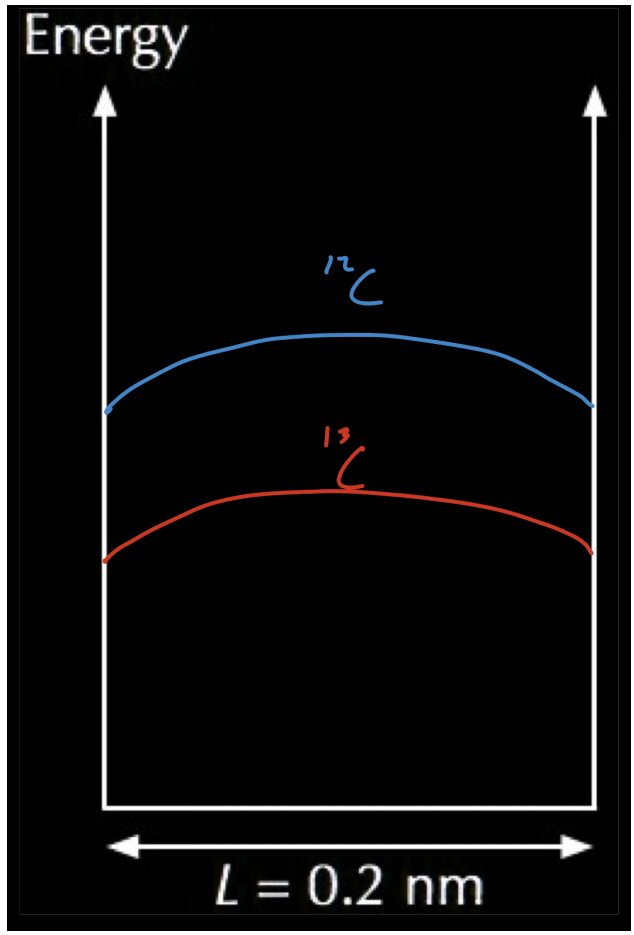

Figure 1: Ground-state ($n = 1$) wavefunctions for ^{12}C (blue) and ^{13}C (red). Because ^{12}C is lighter, it has a higher energy.

$$E_1(^{12}\text{C}) = \frac{(6.626 \times 10^{-34})^2}{8 \times 2.004 \times 10^{-26} \times (2 \times 10^{-10})^2} = 6.85{\times}10^{-23}\ \text{J} = \frac{6.85 \times 10^{-23}}{1.602 \times 10^{-19}}{\times}10^3\ \text{meV} \approx \boxed{0.427\ \text{meV}}$$

$$E_1(^{13}\text{C}) = \frac{(6.626 \times 10^{-34})^2}{8 \times 2.171 \times 10^{-26} \times (2 \times 10^{-10})^2} = 6.32{\times}10^{-23}\ \text{J} = \frac{6.32 \times 10^{-23}}{1.602 \times 10^{-19}}{\times}10^3\ \text{meV} \approx \boxed{0.394\ \text{meV}}$$

(c) Both isotopes have the same transition-state energy, so their activation energies are the delta. Since ^{12}C has a higher ground-state energy, it has a lower activation energy and reacts faster.

$$\frac{\text{rate}(^{12}\text{C})}{\text{rate}(^{13}\text{C})} = e^{(E_1^{12} - E_1^{13})/k_B T}$$

$$\Delta E = 6.85 \times 10^{-23} - 6.32 \times 10^{-23} = 5.3 \times 10^{-24}\ \text{J}$$

$$k_B T = (1.381 \times 10^{-23})(300) = 4.14 \times 10^{-21}\ \text{J}$$

x

$$\frac{\text{rate}(^{12}\text{C})}{\text{rate}(^{13}\text{C})} = e^{5.3\times10^{-24}/4.14\times10^{-21}} = e^{1.28\times10^{-3}} \approx \boxed{1.00128}$$

^{12}C reacts about 0.13% faster.

(d) At higher temperature, thermal energy k_BT grows, so the energy difference ΔE becomes smaller. The ratio approaches 1, so both isotopes react at nearly the same rate.

(e) For ^{13}C to react faster, the transition state energy for ^{13}C would need to be lower than for ^{12}C. This would give ^{13}C a smaller activation barrier even though it starts at a lower ground-state energy, making it the faster-reacting isotope.

3.

(a) Early fatty acid membranes are leaky: small molecules like nucleotides pass through freely, but long RNA polymers cannot. As RNA replicates inside the vesicle, free nucleotides get incorporated into strands and become trapped. The concentration of free nucleotides inside the vesicle drops, so more enter from outside to compensate. New nucleotides flow in, get locked into RNA, and then cannot leave.

(b) RNA polymers accumulating inside the vesicle increase the total solute concentration relative to the outside. Water moves in by osmosis to equalize the concentration difference, and the vesicle swells up as a result.

(c) A swollen, RNA-rich vesicle has its membrane under tension. That tension is relieved when new fatty acid molecules are incorporated, lowering the free energy of the system. So swollen vesicles absorb fatty acids from the surrounding solution and from unstressed, RNA-poor neighbors, which then shrink. Better replication leads to more swelling, more membrane growth, and ultimately more division. This is selection acting at the level of the protocell, driven entirely by physics rather than genetics.

(d) For RNA to replicate, the double strand must first be pulled apart so each strand can serve as a template. Szostak proposes this happened through thermal cycling in volcanic environments: convection currents in pools near hot rocks would periodically carry protocells past a heat source, briefly melting the double strand into single strands. As the protocell returned to cooler water, each strand would template a new complementary copy. No enzymes are required, but just the right physical environment.

4.

(a)

$$\frac{P(L)}{P(D)} = e^{\Delta E/k_BT}$$

$$\Delta E = 10^{-17} \text{ eV}$$

$$k_BT = (8.617 \times 10^{-5})(295) = 0.02542 \text{ eV}$$

$$\frac{P(L)}{P(D)} = e^{10^{-17}/0.02542} = e^{3.93\times10^{-16}} \approx \boxed{1 + 3.93 \times 10^{-16}}$$

The two forms are essentially equally probable at room temperature. The L-form is on the order of 10^{-16} more likely.

(b)

$$\frac{P(D)}{P(L)} = e^{-\Delta E/k_B T} = 0.1$$

$$-\frac{\Delta E}{k_B T} = \ln(0.1) = -\ln(10)$$

$$k_B T = \frac{\Delta E}{\ln 10} = \frac{10^{-17}}{2.303} = 4.34 \times 10^{-18} \text{ eV}$$

$$T = \frac{4.34 \times 10^{-18}}{8.617 \times 10^{-5}} \approx \boxed{5.04 \times 10^{-14} \text{ K}}$$

This is far below anything achievable in nature.

(c) No, the effect at room temperature is only on the order 10^{-16}, which is nowhere near enough to explain how there is near 100% L-amino acid homochirality in biology.

(d) The alternative explanation is that homochirality was a historical accident. Abiotic chemistry produces a 50/50 mix of L and D amino acids, but by chance one form ended up slightly more prevalent early in the origin of life. Once one handedness had a small advantage, it became locked in and amplified over time since mixing L and D disrupts protein function.